



1. Given that $i = \sqrt{-1}$, evaluate the expression $\left(\frac{i+1}{i-1}\right)^2$.

- (A) $-i$ (B) i (C) $-1 - i$ (D) $1 + i$ (E) -1

2. Find the number of integers n such that $\sqrt{n^2 - 24}$ is an integer.

- (A) 0 (B) 2 (C) 4 (D) 7 (E) 8

3. For how many natural numbers $n \leq 12$ is $(n - 1)!$ not divisible by n ?

- (A) 3 (B) 4 (C) 5 (D) 6 (E) 7

4. Bob walks a block east from his house. After each block he walks, he flips a fair coin and walks one block east if the coin is heads and one block west otherwise. He stops his walk the first time he returns home. What is the probability he stops his walk after walking exactly eight blocks?

- (A) $\frac{1}{64}$ (B) $\frac{5}{32}$ (C) $\frac{7}{32}$ (D) $\frac{13}{256}$ (E) $\frac{5}{128}$

5. If each letter in $\left(a + \frac{b}{c}\right)(d + e)$ is replaced by a different integer from 1 to 9 inclusive, what is the minimum integer value of the expression?

- (A) 5 (B) 6 (C) 7 (D) 8 (E) 9

6. Let x and y be positive real numbers such that $\frac{1}{x^2} + \frac{1}{xy} = \frac{1}{9}$ and $\frac{1}{y^2} + \frac{1}{xy} = \frac{1}{16}$. What is the value of $x + y$?

- (A) $\frac{97}{12}$ (B) 9 (C) $\frac{39}{4}$ (D) 10 (E) $\frac{125}{12}$

7. Which of the following could *not* be the discriminant of a quadratic in the form $ax^2 + bx + c$ where $a > 0$ and b, c are integers? (The discriminant of $ax^2 + bx + c$ is $b^2 - 4ac$.)

- (A) 84 (B) 85 (C) 99 (D) 121 (E) 133

8. Alison, Beth, and Carol are each driving the same distance but they own different cars. Alison's car is refueled every 160 miles, Beth's every 240 miles, and Carol's every 400 miles. They all start with full gas tanks, but Alison finishes with an empty tank, Beth finishes with 80 miles worth of gas remaining, and Carol finishes with 160 miles worth of gas remaining. Given that the trip is between 5000 and 6000 miles long, what is the total length of the trip?

- (A) 5000 (B) 5160 (C) 5320 (D) 5400 (E) 5820

9. In quadrilateral $ABCD$ let E be the intersection of AC and BD . If $\angle ACD = \angle CAB = 90^\circ$, $AE = a$, $EC = b$, and the area of $\triangle AED$ is S , which of the following expresses the area of $ABCD$ in terms of S, a, b ?

- (A) $S \cdot \frac{a^2}{b^2}$ (B) $S \cdot \frac{ab}{(a+b)^2}$ (C) $S \cdot \frac{ab}{a^2+b^2}$ (D) $S \cdot \frac{a^2+b^2}{ab}$ (E) $S \cdot \frac{(a+b)^2}{ab}$

10. Alice and Bob are having a picnic. In total, they know 6 other men and 6 other women. How many different sets of friends can they invite so that they invite the same number of men and women to the picnic? (Assume that they invite at least one person.)

- (A) 720 (B) 923 (C) 1023 (D) 1440 (E) 1846

11. Bob has 4 indistinguishable oranges and 13 indistinguishable grapes. He would like to hand out all 17 of his fruit to his 4 students such that each student has more grapes than oranges. How many ways can he do so?

- (A) 840 (B) 1960 (C) 2250 (D) 3024 (E) 5040

12. Two ants are on the xy -plane, one at $(-1, 1)$ and the other at $(2, 0)$. There is an ant-repellent puddle occupying the region $x^2 + y^2 < 1$. What is the length of the shortest path from one ant to the other which does not pass through the puddle?

- (A) $1 + \frac{\pi}{6} + \sqrt{3}$ (B) $\sqrt{10}$ (C) $5 - \sqrt{3}$ (D) $\frac{3\pi}{4} - \sqrt{3}$ (E) 4

13. Let $S(n)$ denote the sum of the digits of n . Find $S(S(k))$ if

$$k = 1! + 2! + 3! + \cdots + 2016! + 2017!.$$

- (A) 24 (B) 25 (C) 26 (D) 27 (E) 28

14. Let $\triangle ABC$ be a triangle such that $\angle C = 90^\circ$, $AB = 137$, and the distance from C to the incenter of the triangle is $28\sqrt{2}$. What is the area of $\triangle ABC$?

- (A) 2870 (B) 3276 (C) 4400 (D) 4620 (E) 5280

15. What is the value of the following sum?

$$\sum_{k=0}^{89} \frac{1}{1 + \tan^3(k^\circ)}.$$

- (A) 45.5 (B) 50 (C) $30\sqrt{3}$ (D) $\frac{181}{4}$ (E) 270

16. In $\triangle ABC$, $\angle ABC = 60^\circ$. Let P be a point inside $\triangle ABC$ such that $\angle APB = \angle BPC = \angle CPA$, with $PA = 8$ and $PC = 6$. What is the length of PB ?

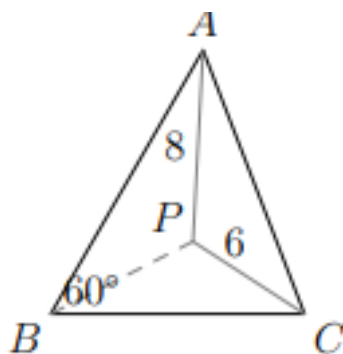


Figure 1: Problem 16

- (A) $3\sqrt{3}$ (B) 5 (C) 6 (D) $4\sqrt{3}$ (E) 7

17. There exists a permutation $a_1, a_2, \dots, a_{16}$ of $\{1, 2, \dots, 16\}$ such that $a_1 = 16$ and whenever $i < j < k$, we have $a_k - a_j \neq a_j - a_i$. Which of these is a possible value of a_5 ?

- (A) 1 (B) 4 (C) 7 (D) 10 (E) 14

18. Let A and B be diametrically opposite points on the circular base of a cone sitting on a flat plane with radius $\frac{3}{4}$ and height $\frac{\sqrt{7}}{4}$. An ant on the point A wants to crawl along the surface of the cone to the point B . Find the minimum distance the ant has to travel.

- (A) $\frac{3}{2}$ (B) $\frac{3\pi}{4}$ (C) $\sqrt{2 + \sqrt{2}}$ (D) $1 + \sqrt{2}$ (E) 2

19. Let $f(x) = 2x\sqrt{1 - x^2}$, and let $f^{(k)}(x)$ denote the k -th iteration of f , that is $f^{(1)}(x) = f(x)$ and $f^{(k+1)}(x) = f(f^{(k)}(x))$ for all positive integers k . For how many distinct real values n is $f^{(2017)}(n) = \frac{1}{2}$?

- (A) 1 (B) 2 (C) 2^{2017} (D) 2^{2018} (E) $2^{2018} + 1$

20. A circle is in the coordinate plane such that the slope of the line from the origin to its center is 2 and the slope of one of its tangents through the origin is 3. Find the slope of the other tangent line through the origin.

- (A) 1 (B) $\frac{13}{9}$ (C) $\frac{17}{12}$ (D) $\frac{2\sqrt{6}}{3}$ (E) $\frac{4\sqrt{17}}{11}$

21. A 20-foot long line of army men are marching in a straight line at a constant speed. A messenger at the back of the line sprints to the front at a constant speed to deliver a message to the captain, then immediately sprints back to the back at the same speed. Once the messenger reaches the back of the line, the army men have traveled a total of 20 feet. How many feet did the messenger sprint?

- (A) 20 (B) $20 + 20\sqrt{2}$ (C) 200 (D) $200 + 100\sqrt{2}$ (E) $40\sqrt{2}$

22. Bob places 3 identical tennis balls in each of 3 different bowls. He then chooses one ball from a random bowl (that has at least one ball in it) and moves it into another random bowl. He repeats this process 4 more times. After he has finished, what is the probability that there are still 3 balls in each bowl?

- (A) $\frac{1}{9}$ (B) $\frac{5}{32}$ (C) $\frac{3}{18}$ (D) $\frac{17}{324}$ (E) $\frac{5}{108}$

23. Alice and Bob live on opposite vertices of a planet in the shape of a cube. They set off at the same time and walk along edges at a rate of one edge per minute. Each time they reach a vertex, they choose one of the 3 neighboring vertices uniformly at random. What is the expected number of minutes it will take for them to meet? (They can meet at the midpoint of an edge.)

(A) 3 (B) 5.5 (C) 8.5 (D) 9 (E) 13

24. Bob is standing at the edge of a cliff. If he moves one unit toward the edge he will fall off. Every minute Bob moves 1 unit away from the edge with probability $\frac{3}{4}$ and moves 1 unit toward the edge with probability $\frac{1}{4}$. What is the probability that Bob falls off the cliff?

(A) $\frac{1}{5}$ (B) $\frac{1}{3}$ (C) $\frac{1}{2}$ (D) $\frac{5}{12}$ (E) 1

25. Bob lists all the powers of 2 from 2^0 to 2^{2017} . He observes that 2^{2017} has leading digit 1, 250 numbers have leading digit 3, and 161 numbers have leading digit 5. Find the count of numbers in his list that have leading digit 1 or 2.

(A) 943 (B) 956 (C) 965 (D) 971 (E) 982

Time limit: 75 minutes